

Adaptive Hybrid Retrieval for Multi-Hop Question Answering: Dynamic Routing Between Dense and Graph-Guided Retrieval, with Vectorless Retrieval as an Exploratory Baseline

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Abstract—Retrieval-augmented generation (RAG) has emerged as an effective paradigm for grounding large language models in external evidence, yet many existing systems still rely on a fixed retrieval strategy. This design becomes restrictive in multi-hop question answering, where answering a single query may require semantic matching, relational expansion, and navigation across distributed evidence. Dense retrieval is efficient and widely used, but it may fail to preserve multi-step evidence chains. Graph-guided retrieval captures relations among entities and passages more explicitly, although its benefits are not uniform across all query types and it introduces additional indexing overhead. Structure-aware retrieval methods further suggest that hierarchical document organization can improve evidence access in long-context settings. Motivated by these complementary strengths and limitations, this paper investigates an adaptive hybrid retrieval framework for multi-hop question answering. The proposed formulation dynamically routes queries between dense retrieval, graph-guided retrieval, and a vectorless structural branch treated as an exploratory baseline. The study positions this framework within recent advances in adaptive retrieval, graph-guided evidence expansion, and structure-aware retrieval and argues that query-dependent routing offers a principled direction for improving evidence selection in multi-hop question answering.

Index Terms—retrieval-augmented generation, multi-hop question answering, adaptive routing, graph-guided retrieval, dense retrieval

I. INTRODUCTION

Retrieval-augmented generation (RAG) improves knowledge-intensive question answering by combining a language model with an external retriever rather than relying solely on parametric memory. The retrieval module is therefore central to system quality, particularly when the answer cannot be inferred from a single passage but must instead be assembled from several dependent pieces of evidence [1], [2]. This challenge is especially visible in multi-hop question answering, where relevant facts are often distributed across documents and must be connected through intermediate reasoning steps [3], [4].

A major limitation of many current RAG pipelines is their dependence on a single retrieval paradigm. Dense retrieval

remains the dominant choice because it is simple, scalable, and effective for semantic matching. However, multi-hop question answering frequently requires evidence gathering that is conditional and sequential rather than purely similarity-based [2]. MultiHop-RAG demonstrates that conventional retrieve-then-generate pipelines degrade when support must be drawn from multiple documents and combined through multi-step reasoning [3]. Even when relevant passages are retrieved, the presence of noisy or only partially relevant evidence can still reduce downstream answer quality [5].

Graph-guided retrieval provides a different retrieval bias by explicitly modeling relations among entities, chunks, or documents. HippoRAG shows that graph-structured memory can support multi-step retrieval through associative propagation, while KG²RAG starts from semantic seed retrieval and then expands evidence using knowledge graph relations to improve coherence and diversity [6], [7]. HybGRAG further demonstrates that some questions require both textual and relational evidence, which suggests that neither dense retrieval nor graph retrieval alone is universally sufficient [8]. Comparative analysis also indicates that dense RAG and GraphRAG exhibit complementary strengths rather than a single dominant behavior across tasks [9].

A third perspective emerges from structure-aware retrieval. RAPTOR improves retrieval over long documents by organizing information into a hierarchical summary tree rather than relying only on flat chunk matching [10]. Edge *et al.* likewise show that graph-based organization can support local and global evidence access for query-focused summarization [11]. Although vectorless retrieval remains less established in the peer-reviewed literature than dense and graph-guided methods, these structure-oriented approaches suggest that hierarchical document organization may provide an alternative retrieval signal for complex or long-context queries.

The motivation of this paper is therefore to investigate whether retrieval itself should be adaptive. Rather than assuming that one retriever is optimal for all questions, the paper considers a hybrid framework in which dense retrieval

addresses local semantic queries, graph-guided retrieval addresses relation-heavy multi-hop queries, and a vectorless structural branch is treated as an exploratory baseline for structure-sensitive questions. The objective is to establish the literature foundation for a routed retrieval design that dynamically selects among these paradigms for multi-hop question answering [1], [7], [8], [10].

II. RELATED WORK

The literature on multi-hop question answering consistently shows that one-shot retrieval is poorly matched to questions whose supporting facts must be discovered and combined across several reasoning steps. IRCoT addresses this mismatch by interleaving retrieval with chain-of-thought reasoning so that intermediate inference can guide subsequent evidence acquisition [2]. MultiHop-RAG extends this concern from system design to evaluation by constructing queries whose evidence is intentionally distributed across documents, thereby exposing weaknesses in standard RAG pipelines [3]. HotpotQA remains important in this setting because it provides supporting-fact supervision and therefore makes retrieval quality observable rather than purely implicit in the final answer [4]. Together, these works establish that multi-hop QA requires evidence composition rather than simple nearest-neighbor retrieval.

Graph-based retrieval methods address this issue by making relationships among evidence units explicit. HippoRAG models long-term knowledge through a graph-like memory structure that improves multi-step retrieval via concept associations [6]. KG²RAG similarly begins with semantic retrieval but then expands and organizes evidence using knowledge graph relations, improving coherence for downstream generation [7]. HybGRAG sharpens this argument by showing that hybrid questions over semi-structured knowledge bases often require both relational traversal and textual grounding, which neither dense retrieval nor graph retrieval can provide alone [8]. Han *et al.* reinforce this conclusion through systematic comparison, reporting that dense RAG remains competitive on detail-oriented tasks while GraphRAG-style approaches are more advantageous when broader relational aggregation is required [9]. The implication is not that graph-guided retrieval should replace dense retrieval, but that different query types benefit from different retrieval biases.

Structure-aware retrieval contributes a complementary line of evidence. RAPTOR demonstrates that hierarchical abstraction can improve retrieval over long documents by retrieving across multiple levels of a summary tree rather than only over flat chunks [10]. Edge *et al.* similarly show that graph-based corpus organization can support both local and global evidence access in query-focused retrieval settings [11]. These methods do not constitute a mature vectorless literature in the same sense as dense or graph-guided retrieval, yet they strongly suggest that document structure can function as an independent retrieval signal. This observation motivates the exploratory vectorless branch in the present topic, even though

the primary scholarly foundation remains dense, graph-guided, and adaptive retrieval research.

The adaptive dimension is most directly represented by Adaptive-RAG, which formulates retrieval selection as a query-complexity decision and routes questions to strategies of differing cost and sophistication [1]. The broader survey literature on agentic RAG further indicates that retrieval pipelines are increasingly being treated as controllable processes involving planning and dynamic strategy selection rather than static modules [12]. However, much of this work remains either architectural or survey-oriented. A narrower empirical gap remains at the intersection of multi-hop QA and hybrid retrieval: most existing studies optimize one retrieval paradigm at a time, while the accumulated evidence suggests that different questions favor dense, graph-guided, or structure-oriented retrieval for different reasons. This gap motivates the proposed direction of adaptive hybrid retrieval, where routing becomes the central mechanism for deciding which retrieval paradigm should be applied to a given multi-hop query.

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